



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST

AIR LEAK TESTER

MODEL LS-1822A

ERROR AND TROUBLESHOOTING MANUAL

1. ERRORS

If an error occurs when the power is turned on or during K(Ve) calibration, a leak test, or a Mastering process, an error message appears on the LCD, an LED indicator blinks and an error signal is transmitted through the control I/O port.

Errors indicated only by LED's blinking

- ROM check and RAM check If the ROM or the RAM is defective, the Hi NO GO and LO NO GO LED light up. No operations are possible.
Contact a Cosmo representative.

1.1 Error messages and associated countermeasures

Code	Message	Probable cause and countermeasures
1	Exceeding M.COMP Limit	The Mastering value has exceeded M.COMP limits. Extend the CHG and the BAL1 timers. The leak rate during Loop #1 will be smaller, which allows the resetting of a smaller Mastering value.
2	A/D Converter Mal-function!	Contact a Cosmo representative to have the A/D converter replaced with a new one.
3	DP SENSOR Mal-function!	The DPS power supply or electric circuit may be malfunctioning, or a cable may be disconnected. Contact a Cosmo representative.
5	DPS Offset Exceeding Limit	The DPS offset is out of Tolerance when the power is turned on. Adjust the DPS offset in the CALIB mode.
6	PS Offset Exceeding Limit	The PS offset is out of Tolerance when the power is turned on. Adjust the PS offset in the CALIB mode.
8	Improper Test Pressure	The test pressure exceeds the programmed test pressure limits at the end of the CHG stage due to inappropriate P.Lo (P.Hi for vacuum) settings, or to PS malfunction.
9	DPS Output saturation	The DPS output during the BAL1, BAL2 and DET stages is out of the allowable range. Check the DPS offset.
10	PS output saturated	The PS output during leak testing is out of the allowable range. Check the PS offset and the test pressure setting. See Section 13.7.
11	K(Ve) out of range	The measured K(Ve) exceeds 100L. Check the leak rate of the Leak Master.
12	Leak Limit out of range	Leak rate limits exceed the allowable range that is calculated based on the measured K(Ve) (program limitation). Perform calibration after changing the unit to a pressure unit. See Section 8.1.2. Or set lower leak limits.
15	Air Valve Inactive C1	The equalization valve AV2, exhaust valve AV3, and/or solenoid valves for pilot pressure are/is malfunctioning. Contact a Cosmo representative.

Code	Message	Probable cause and countermeasures
	Air Valve Inactive C2	Fill valve AV1 and/or the solenoid valve for pilot pressure are/is inactive, or the test pressure may be insufficient. Verify the test pressure and contact a Cosmo representative if necessary.
	Air Valve Inactive C4	The DPS sensitivity may have been lowered. Check the sensitivity. If the test pressure is low, the BLW timer setting may be too short. Verify the BLW timer setting and contact a Cosmo representative if necessary.
	Air Valve Inactive C5	For H15, equalization valve AV2 is inactive during the BAL2 stage. For the Y or C circuit, balance valve AV1 is inactive during the BAL2 stage. Apply the appropriate pilot pressure (except for the H15 circuit and the C circuit). Verify pilot pressure and contact a Cosmo representative if necessary.
17	P.Lo(Hi) Value is 0 (zero)	Test pressure limit, P.Lo (P.Hi for vacuum), is set to 000.000.
18	RTC Battery discharge	The built-in battery has been completely discharged. Contact a Cosmo representative.

NOTE: Error codes 4, 7, 13, 14 and 16 are currently not used.